Toward Hierarchical Adaptive Alignment for Aerial Object Detection in Remote Sensing Images

Chenwei Deng, Donglin Jing, Yuqi Han, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— The aerial objects tend to distribute with a major variation in the scale and arbitrary orientations in remote sensing images. To meet such characteristics of the aerial object, most of the existing anchor-based detectors rely on preset anchors with variable scales, angles, and aspect ratios, which lead to the misalignment of the selection of candidate regions, the extraction of object features, and the label assignment of preset boxes, interfering the performance of the detector. To address this issue, we propose a hierarchical adaptive alignment network (HAA-Net). Specifically, we first design the region refinement module (RRM), feature alignment module (FAM), and potential label assignment module (PLAM) to alleviate the misalignment of the region, feature, and label levels, respectively; furthermore, we use the gradient equalization strategy to jointly optimize these modules at different levels, so that the whole network can be fully trained to significantly improve detection performance. Extensive experiments demonstrate that our approach can achieve superior performance in three common aerial object datasets [(e.g., Dataset of Object Detection in Aerial Images (DOTA), High-Resolution Ship Collection 2016 (HRSC2016), and University of Chinese Academy of Sciences-Aerial Object Detection (UCAS-AOD)] when compared with state-of-the-art detectors.**

***Index Terms*— Aerial object detection, dynamic anchor selection, feature alignment, region alignment.**

I. Introduction

As FOR aerial images, object detection is a significant technology of computer vision applied for object recognition and location, which has been widely employed in several fields (e.g., crop monitoring, resource exploration, as well as environmental protection). With the rapid growth of remote sensing (RS) data, it progressively becomes a challenge to extract objects of interest from a mass of RS images. Conventional methods use hand-made feature operators and sliding windows to attain the characteristic of texture, edge, and so on. Nevertheless, those methods of manual modeling for feature extraction [1], [2], [3], [4], [5] have been confronted with setbacks when attempting to effectively distinguish the object from the cluttered backgrounds, which share similar texture or shape with the targets. As a result, these methods often suffer from insufficient robustness and poor performance in port [6] and other complex scenarios [7], [8], [9], [10], [11].

Over the past few years, the performance of object detection has made remarkable progress benefiting from the intensive study of convolution neural networks (CNNs). Existing frameworks of CNN-based detection [12], [13], [14], [15] could be primarily divided into two categories, including one-stage and two-stage detectors [16], [17]. For one stage, they generally preset several horizontal anchors on the feature maps after extracting features by convolution. Then, based on these anchors, regression and classification operations are conducted to obtain the target boundary. However, in RS images, targets (e.g., ships and aircraft) are often distributed with different scales, aspect ratios, and directions. In this case, the misalignment caused by horizontal anchors in mainstream one-stage detectors would generate severe background disturbance and mislead detection when predicting rotated targets in RS images. In order to eliminate the misalignment of the candidate region, gliding vertex [18] avoided the interference of the horizontal candidate region by regressing the coordinates of the four vertices of the detection box. R^3Det [19] used horizontal anchors to increase speed and generate more candidate boxes, and then used the rotated anchors during the refining stage to accommodate scenarios where the target is dense and distributed with arbitrary orientations. Furthermore, some methods would preset extensive rotation anchors [20], [21], [22] to locate targets with arbitrary orientations. However, such an operation could not only cause redundant calculations but also could not guarantee accurate features for classification and regression tasks.

For achieving higher detection performance, the advanced two-stage target detectors [12], [23] were proposed. This detection framework is generally composed of two parts: a detection head based on R-CNN and a regional candidate network (RPN). First, RPN is utilized to produce high-quality regions of interest (RoI). Subsequently, RoI pooling can acquire precise features in accordance with RoI. Finally, classifiers and regressors are used again to refine the bounding box for precision detection. Compared with one-stage detectors, RoI improves the quality of candidates and achieves better performance. However, the horizontal RoIs can lead to a sort of misalignment between the bounding box and the rotated target as well as a higher computational complexity. To this end, for detecting the rotated target, SCRDet [24] developed a supervised multidimensional attention network (MDA-Net) composed of pixel attention network and channel attention

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network to reduce the adverse impact of background noise in the misaligned candidate area. R^2CNN [25] used a rotated RoI pool layer, which makes the convolution kernel extract pooled features of RoI with different pooled sizes and ensures accurate detection. However, these methods introduce more complex convolution structures and result in computational redundancy, which is difficult to effectively extend to other RS detectors and hardware platforms.

Through the analysis earlier, we can see that an important problem that the existing RS detector still faces is the serious misalignment in the different modules of the RS detector. We call this phenomenon hierarchical misalignment. This phenomenon can lead to gradually accumulating the error of the network, which makes it difficult for the detector to adaptively adjust the direction of network fitting according to the real target, thus affecting the performance of the detector. In our view, the remaining hierarchical misalignment issues for the existing detectors in the RS detection field can be categorized into the following three aspects.

A. Region Level Misalignment

The region covered by anchors is misaligned with the ground truth (GT). To be specific, we illustrate the detection results of the advanced detector RetinaNet [26] in challenging scenarios for better understanding. As shown in Fig. 1(a), for covering the target, the number of preset rotated anchors often increases linearly with angular spacing becoming dense, which results in considerable redundant calculations and severe background interference. This kind of phenomenon is particularly obvious in places, such as airports and ports, which contain dense targets.

B. Feature Level Misalignment

The convolution kernel with a fixed receptive field horizontally sliding is difficult to align with features for rotated targets. As shown in Fig. 1(b), the axis-aligned convolution kernel cannot accurately extract the target features in the preset anchor due to the multidirectional distribution of the target.

C. Label Level Misalignment

The existing method employs intersection over union (IoU) as a label assignment standard, which could hardly determine the quality of the preset anchors. To this end, they may suffer from the misalignment between labels and samples. As can be seen from Fig. 1(c), the subfigure on the left-hand side indicates that anchors with high IoU cannot ensure the effectiveness of detection, while the right-hand side shows that an anchor with small IoU still has the potential to localize objects accurately since it captures the most distinguishing features for the target.

To sum up, a natural concern comes that how to achieve systematic hierarchical alignment. This can not only reduce the computational complexity but also improve the accuracy of the model. To this end, we propose HAA-Net as a solution. Specifically, HAA-Net is proposed with three modules, including the region refinement module (RRM), feature alignment module (FAM), and potential label assignment module (PLAM). In RRM, we use parallel regression branches to refine the direction and position of the preset anchor for the generation of high-quality rotated boxes. Then, rotated convolution in FAM is utilized to adaptively align the convolution features following the shape, size, and direction of the corresponding anchor. Next, PLAM is employed to set the high-quality anchor capturing the key feature as the positive samples and eliminating useless samples. Finally, during the training process, high-quality anchors selected by PLAM would participate in the calculation of the loss function. The detector can use these high-quality anchors and corresponding feature maps as guidance, and further optimize the convolution parameters in RRM and FAM by using back propagation so that each module of the detector can be fully trained. The main contributions are summarized as follows.

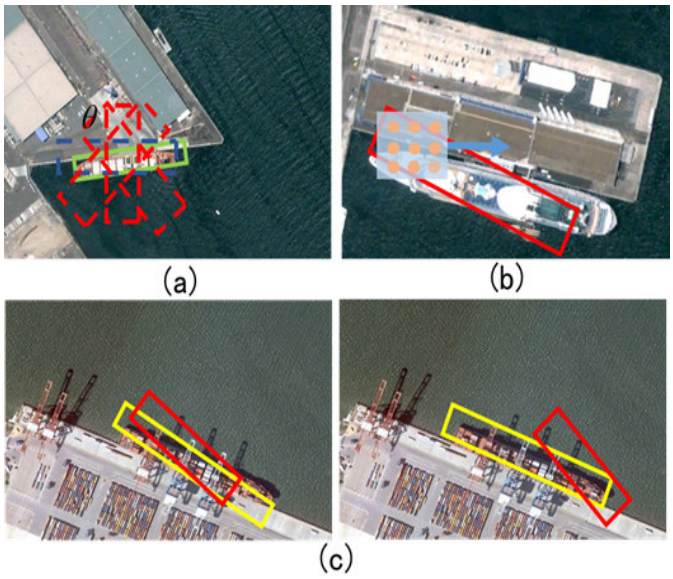


Fig. 1. Remaining the misalignment issues for target in RS images. (a) Misalignment between the anchor covering targets and the corresponding GT (represented by red anchor and green GT). (b) Misalignment between the axis-aligned convolution (represented by a light blue rectangle, orange dots mean the value of convolution kernels) and an anchor box. (c) Misalignment between labels and samples (represented by red anchor and yellow regression box).

1) We comprehensively analyze the network misalignment issue that limits the performance of the detectors.
2) We propose an adaptive alignment detection framework, to alleviate the misalignment of the region, feature, and label levels, respectively.
3) We design a joint optimization strategy to make the training of the whole network more balance so that higher performance can be achieved.

The remainder of this article is organized as follows. Section II reviews the related works and analyzes the existing problems. In Section III, a detailed description of the proposed HAA-Net is presented. In Section IV, extensive experiments are conducted and the results are discussed. Conclusions are summarized in Section V.

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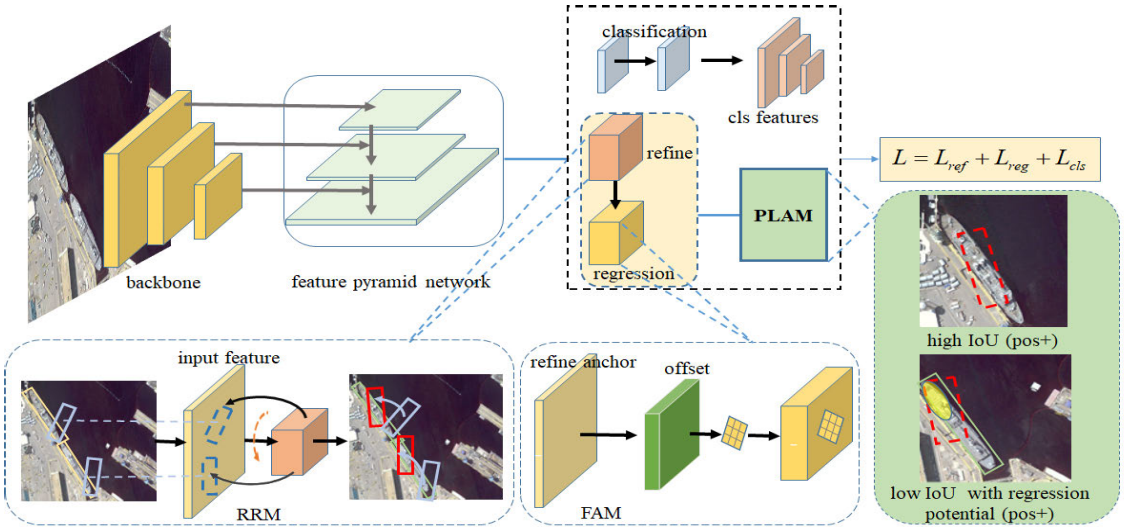


Fig. 2. Framework of the proposed HAA-Net.

II. Related Works

In this section, the major works in region alignment are first reviewed, and then some representative works on feature alignment and label alignment are also introduced. However, it should be noted that most of these works focus on the partial misalignment phenomenon of different levels.

A. Region Alignment

The existing detector based on CNN in RS images is often migrated from the framework of natural scenes. Some methods [4], [27], [28] were directly migrated from the general horizontal detection framework. These horizontal anchors or regions are difficult to distinguish aerial targets distributed in any direction and are prone to misjudgment due to background interference. To solve this problem, some researchers have introduced some additional direction dimensions to realize multidirectional target detection. For example, R^2PN [29] was proposed to detect a rotated ship by generating a multiorientated RoI. R^2CNN [30] used a rotated RoI pool layer, which makes the convolution kernel extract pooled features of RoI with different pooled sizes and ensures accurate detection. And some works [12] integrated angle information into bounding box regression of YOLOv2 [13] to detect rotated ships.

However, for achieving a higher overlap with the aerial target, these methods present densely rotated anchors (or rotated regions). Due to the small overlap area between most anchors and targets, it brings great redundant calculation and serious background interference. RoI transformer [31] based on spatial transformation was proposed to convert horizontal RoI into rotated one, to avoid presetting many anchors. But it still needs to preset heuristically defined anchor and complex RoI transformation operations. Different from the previous works, our proposed RRM can adaptively modify the anchor according to the target spatial distribution, and automatically align with the GT.

B. Feature Alignment

Some works used RoI operators (such as RoI pooling [32] and deformable RoI pooling [33]) to extract features in RoIs. These features can approximately represent the spatial information of the target. The RoI pooling operation first divides the RoI into subregion grids and then takes the maximum output of each subregion as the corresponding output grid unit. Deformable RoI pooling adds an offset to realize adaptive feature selection of target sampling points, capturing the feature of the target boundary. However, these RoI operators usually involve a large number of regional operations, such as feature distortion and feature interpolation, which greatly reduces the detection speed of the network.

Manually designed features operator or algorithms cannot cope with overly complex deformations. Therefore, deformable convolution (DC) can add an offset to change the spatial sampling position and learn the offset from the different detection tasks without additional manual adjustment. These new modules can be plug-and-play, easily replacing the ordinary modules of the existing CNN for end-to-end training. On this basis, guide anchor [32] used the shape of the anchor as important information to align with features. It learns the offset from the prediction box after anchor regression and then guides DC to realize feature alignment. Moreover, Aligndet [34] designed RoI convolution, achieving the same effect with RoI alignment in a single-stage detector. The application of these methods often leads to poor performance in the detection of adjacent targets in aerial images. Different from the above-mentioned methods, the proposed method adjusts the feature sampling position through the guidance of the anchor to reduce the misalignment between the axially aligned convolution feature and the target in any direction.

C. Label Alignment

1) Hard Label Assignment: Marking each anchor as a positive (pos) or negative (neg) sample is the key to training the network. Most anchor-based detectors intensively preset

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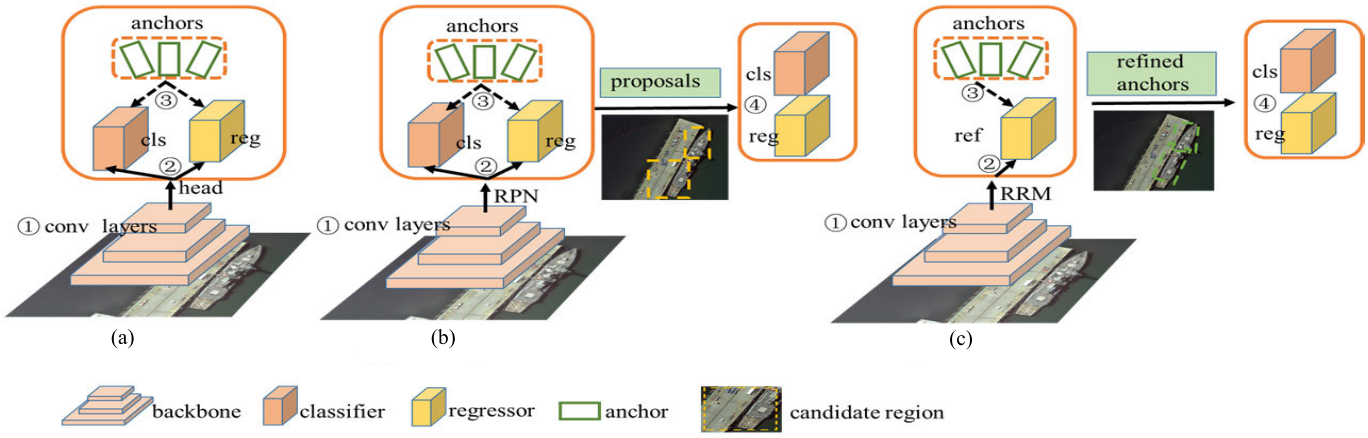


Fig. 3. Comparison of the different detection frameworks. (a) One-stage detector. (b) Two-stage detector. (c) Our HAA-Net.

anchors and set the label of the anchor by calculating the IoU with GT. A large number of preset anchors lead to serious imbalance problems, especially for arbitrarily oriented objects with additional angle settings. The most common solution is to control the proportion of candidates through specific sampling strategies [35]. Some advanced strategies [36] have been proposed to dynamically select pos samples for each GT. Adaptive training sample selection (ATSS) [37] selected top-k anchors from each layer of the feature pyramid and used the mean and variance value of these top anchors as the pos/neg segmentation threshold. Probabilistic anchor assignment (PAA) [38] was proposed to adaptively divide anchors into pos/neg anchors in a probabilistic manner based on the joint state of classification and regression losses. Optimal transport assignment (OTA) [39] dealt with the label assignment problem by describing the process as an optimal transportation problem. Hard label assignment treats all samples equally, but it is incompatible with the evaluation indicators in target detection.

2) Soft Label Assignment: Because the prediction box has different quality in the evaluation, the samples should be treated differently during the training. Many works [40], [41] have been proposed to solve the unequal problem of training samples. Focal loss [26] added a modulation factor to the cross-entropy loss to reduce the weight of the loss allocated to well-classified samples, making the detector focus on hard samples. The generalized focal loss [42] assigned a soft weight to each anchor by jointly considering classification score and positioning quality. Varifocal loss [43] used IoU-aware labels as a joint representation of classification and regression to train the network. Most of the above-mentioned methods focus on calculating the weight of pos (wpos) and ignore the regression ability of negative samples. In this article, we excavate potential negative samples with regression ability and design different weights of pos/neg for each anchor respectively.

III. Proposed Methodology

Fig. 2 illustrates the overall pipeline of the proposed HAA-Net framework, and one can see that, unlike the partial alignment approach that existed in the aforementioned methods, we systematically analyze the hierarchical misalignment phenomenon of the RS detectors and develop an adaptive alignment strategy, which is expected to achieve the whole alignment of different levels. Specifically, in order to generate a high-quality rotation anchor, the RRM automatically refines the size and position of the anchor to achieve region alignment. Then, the modified anchor and feature map are sent to the FAM. The alignment convolution in FAM adaptively aligns features according to the target orientation and the corresponding anchor offset. Next, PLAM assigns high-quality anchors that effectively extract target features as positive samples to ensure stability in the training process. Finally, we appropriately suppress the large gradient in the training and enhance the gradient response of high-quality samples to ensure that each module of the network can be trained in a balanced way.

A. Region Refinement Module

The target in the aerial image has the characteristics of large-scale difference and multidirection. The preset anchor often contains the target and many backgrounds, so it is difficult to align with the target. In addition, due to the misalignment between the anchor and target, subsequent feature extraction, label assignment, and other operations would be difficult to work effectively, which can damage the function of each module of the whole network. Therefore, we propose an RRM to provide a high-quality preset anchor by correcting the orientation and position of the anchors, so that they are aligned with GT as much as possible. In this way, we can achieve the adaptive alignment between the anchor and target without a heuristical design. The difference between our method, one-stage, and two-stage is shown in Fig. 3.

Generally, using rectangular bounding boxes to regress multidirectional targets (such as ships) can lead to redundant background noise and unnecessary overlap [44], [45]. Therefore, we adopt a rotated rectangular anchor to match the target in any direction. We use the 1-D direction vector $d = (x, y, w, h, \theta)$ to eliminate the divergence between the bounding box b and the GT g, the specific expressions are as follows:

$$\begin{aligned} d_x &= 1/w_b \cdot (x_b - x_g), \quad d_y = 1/h_b \cdot (y_b - y_g) \\ d_w &= \log w_g - \log w_b, \quad d_h = \log h_g - \log h_b \\ d_\theta &= \tan(\theta_g - \theta_b). \end{aligned} \tag{1}$$

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The original anchors with fixed aspect ratios and scales are not likely to be the best designs. We consider that the critical part of the proposed scheme in two-stage is that the proposals are selected by learning and reduce the search space that covers targets. Meanwhile, it improves the quality of candidates, achieving better performance than the one-stage detector. Given this finding, we intend to integrate this mechanism into the one-stage detectors. We can only regress the direction offset vector $\Delta_{\text{loc}} = (d_x, d_y, d_w, d_h, d_\theta)$ so that we can keep the anchor aligned with the corresponding target. We use p and t to denote the detection result and its corresponding GT. Then, the loss function of the dynamic refine module is defined as

$$L_{\text{ref}} = \text{smooth}_{L_1}\left(\Delta_{\text{loc_}t} - \Delta_{\text{loc_}p}\right) \tag{2}$$

where function smooth_{L_1} is expressed as

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5\ x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{3}$$

where x is the difference between the regression box and the corresponding GT. Compared with other anchor-based detectors, we can only set fewer anchors for each feature map. Moreover, impacted by the automatic refinement of preset anchors, it is not required to carefully set various superparameters (e.g., angles and scales). Consequently, our methods help improve the detection performance of HAA-Net, and the computation is simplified.

B. Feature Alignment Module

Generally, the convolution features extracted from the detector backbone are aligned with the fixed receptive field axis. Specifically, in standard 2-D convolution, we use the sliding window $\text{SW} = \{(x_i, y_j) \mid x_i, y_j = -1, 0, 1, \ldots\}$ to sample the feature map $X_{h\times w}$. However, the aerial targets are always distributed in any direction with a different appearance. Even if an anchor is assigned to a positive sample with a high IoU, there is still a deviation between the box and the convolution feature extracted by the axis-aligned convolution.

DC shown in Fig. 4(a) learns the offset based on the input feature map and then changes the position of the sampling points according to the offset in the convolution process to sample the adjacent positions. However, since the detection task only provides the category and location of the target, the monitoring information for training the offset of the whole feature map is weak in DC. This would lead to the instability of training, making the sampling point offset too large, seriously deviating from the target and sampling to the background. In Fig. 4(a), orange dots represent the sampling location of standard convolution and yellow dots represent the sampling location of DC. We can see that the sampling position of DC deviates from the ship, and the sampling location is in the background (e.g., seawater or shore).

To solve these problems, in FAM, we use the width and height of the anchor to stretch the sampling position of the convolution kernel proportionally and then use the rotation angle of the anchor to rotate the sampling position of the convolution kernel at the same angle, so that the convolution sampling points can fully cover the target without serious deviation. As shown in Fig. 4(b), orange dots represent the sampling position of standard convolution and red dots represent the sampling position of FAM. Compared with DC, using anchor information as guidance, our sampling position can effectively gather near the target, thus achieving accurate extraction of target features.

Specifically, we use the rotated convolution guided by the anchor in FAM to adaptively align the target with its corresponding anchor's offsets. For 2-D rotation matrix $R(\theta) = (\cos\theta, -\sin\theta; \sin\theta, \cos\theta)^T$, when a vector $\mathbf{a}$ is multiplied by $R(\theta)$, only the spatial direction of the vector is changed. Using this property, we consider the sample point $\mathbf{sw}$ of the sliding window as a vector and then rotate it. Additionally, the angle θ between the anchor and the horizontal direction, width, and height is preset. Therefore, as shown in Figs. 5 and 6, $\mathbf{sw}$ can be scaled before rotating, which can adaptively modify the sampling position according to the orientation and size of the anchor. Specifically, inspired by the RoI pooling operation, we divided the anchor coverage into some regions for sampling (k is the convolution kernel size). Then, we multiply $\mathbf{sw}$ by $R(\theta)$ and let $\mathbf{sw}$ rotate in space. And our redefined rotated sample points $\mathbf{sw_r}$ are as follows:

$$\mathbf{sw_r} = ((c_x, c_y) + (w, h) \cdot \mathbf{sw}/k) R^T(\theta) \tag{4}$$

where (c_x, c_y) represents the center of the horizontal anchor for input feature maps on location $\mathbf{m}$. The output in standard 2-D convolution can be defined as:

$$\mathbf{Y}(\mathbf{m}) = \sum_{\mathbf{sw}\in SW} \mathbf{X}(\mathbf{m} + \mathbf{sw}) \cdot \mathbf{W}(\mathbf{sw}) \tag{5}$$

where $\mathbf{W}(\mathbf{sw})$ represents the weight value of the convolution kernel at $\mathbf{sw}$. For each position $\mathbf{m}$, position offset $\mathbf{sw_r} - \mathbf{m}$ is determined by the position of the anchor and convolution kernel sampling locations. The modified output feature map can be defined as

$$\mathbf{Y_r}(\mathbf{m}) = \sum_{\mathbf{sw}\in \text{SW}} \mathbf{X}(\mathbf{sw_r}) \cdot \mathbf{W}(\mathbf{sw}). \tag{6}$$

On that basis, the standard convolution can be modified to the rotational convolution corresponding to the location of the anchor at any position on the input feature map. The architecture of FAM is shown in Fig. 7 With the use of the embedded convolution, we can see that FAM is a lightweight convolution layer, and the velocity delay caused by offset calculation is negligible.

C. Potential Label Assignment Module

In object detection, the features of object recognition and location are not regularly distributed in the same position of the object, especially for those objects with arbitrary orientations and multiple aspect ratios. Therefore, the strategy of label assignment based on the input IoU value cannot capture the useful regression features required. A natural method aims to use the feedback of regression results to represent the matching ability and dynamically guide the training of the model. However, if the output IoU based on training samples is used as a soft label for classification, it would make the

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TABLE I
PERFORMANCE EVALUATION ON THE DOTA DATASET

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Two-stage:																	
R^2CNN [25]	ResNet101	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [48]	ResNet101	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
FR-O [23]	ResNet101	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
RoI-Trans [31]	ResNet101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
Gliding Vertex [18]	ResNet101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
A2RMNet [49]	ResNet101	89.84	83.39	60.06	73.46	79.25	83.07	87.88	90.90	87.02	87.35	60.74	69.05	79.88	79.74	65.17	78.45
MASK-OBB [50]	ResNet101	89.69	87.07	58.51	72.04	78.21	71.47	85.20	89.55	84.71	86.76	54.38	70.21	78.98	77.46	70.40	76.98
SCRDet [24]	ResNet101	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
SCRDet++ [51]	ResNet101	90.01	82.32	61.94	68.62	69.62	81.17	78.83	90.86	86.32	85.10	65.10	61.12	77.69	80.68	64.25	76.24
One-stage:																	
R^3Det [19]	ResNet101	89.54	81.99	48.46	62.52	70.48	74.29	77.54	90.80	81.39	83.54	61.97	59.82	65.44	67.46	60.05	71.69
CADNet [52]	ResNet101	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.9	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
BBAVectors [53]	ResNet101	88.63	84.06	52.13	69.56	78.26	80.40	88.06	90.87	87.23	86.39	56.11	65.62	67.10	72.08	63.96	75.36
R-RetinaNet [26]	ResNet101	88.82	81.74	44.44	65.72	67.11	55.82	72.77	90.55	82.83	76.30	54.19	63.64	63.71	69.73	53.37	68.72
RSDet [54]	ResNet101	89.80	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63,90	65.60	67.20	68.00	72.20
CSL [55]	ResNet152	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
FFA3 [56]	ResNet101	88.80	74.40	48.90	57.90	63.60	75.90	79.60	90.80	80.30	82.90	54.30	60.00	66.90	66.80	42.50	68.90
NPMMR-Det [57]	Darknet53	89.44	83.18	54.50	66.10	76.93	84.08	88.25	90.87	88.29	86.32	49.95	68.16	79.61	79.51	57.26	76.16
GGHL [58]	DarkNet53	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.40	73.38	68.45	70.14	76.95
RIDet-O [59]	ResNet101	88.94	78.45	46.87	72.63	77.63	80.68	88.18	90.55	81.33	83.61	64.85	63.72	73.09	73.13	56.87	74.70
HAA-Net	ResNet101	**90.43**	86.83	**62.23**	**84.71**	71.16	80.61	87.04	**90.90**	**90.50**	76.87	**69.77**	**84.54**	78.60	66.79	70.02	**79.40**

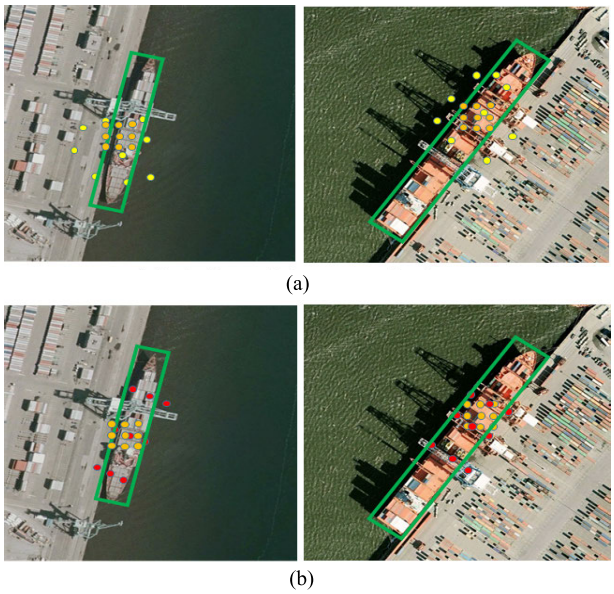


Fig. 4. Illustration of offset parts in the different convolution. (a) DC. (b) FAM (ours).

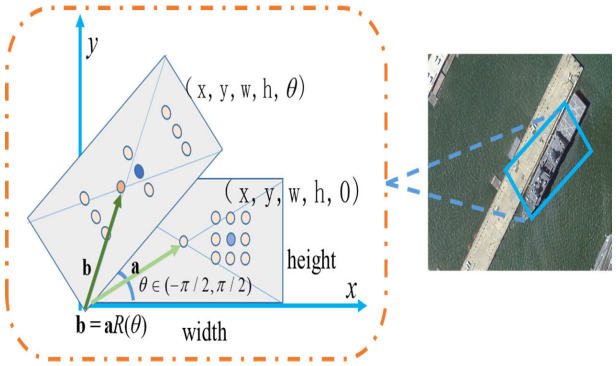


Fig. 5. Diagram of feature alignment. We realize equal interval uniform sampling according to the information provided by the anchor.

TABLE II
EFFECTS OF EACH COMPONENT OF HAA-NET ON HRSC2016

with RRM ?	with FAM ?	with PLAM ?	mAP
✗	✗	✗	83.3%
✓	✗	✗	87.4%
✗	✓	✗	85.3%
✗	✗	✓	87.2%
✗	✓	✓	88.7%
✓	✓	✓	**90.4%**

model difficult to converge due to the accidental high IoU obtained by insufficient training.

According to the analysis earlier, the uncertainty of regression would interfere with the reliability of output IoU during feature alignment. To be specific, high-quality samples may be ignored, and the false-positive samples can lead to the instability of training. Based on the above-mentioned analysis, this study introduces the concept of alignment degree (ad) to replace IoU and uses the prior information of spatial matching, the ability of feature alignment, and the uncertainty of regression during the training to measure the regression performance of a bounding box. It is defined in the following:

$$\mathrm{ad} = \varepsilon \cdot I_{\mathrm{in}} + (1-\varepsilon)\cdot I_{\mathrm{out}} - \mu\sigma(I_{\mathrm{out}} - I_{\mathrm{in}}) \tag{7}$$

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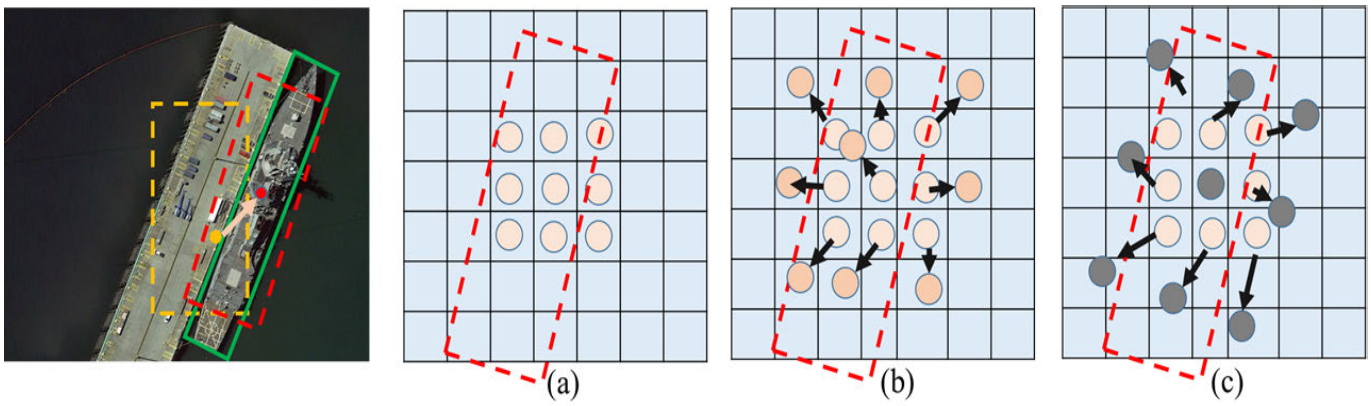


Fig. 6. Comparison of the different convolution kernels. (a) Standard convolution. (b) DC. (c) FAM.

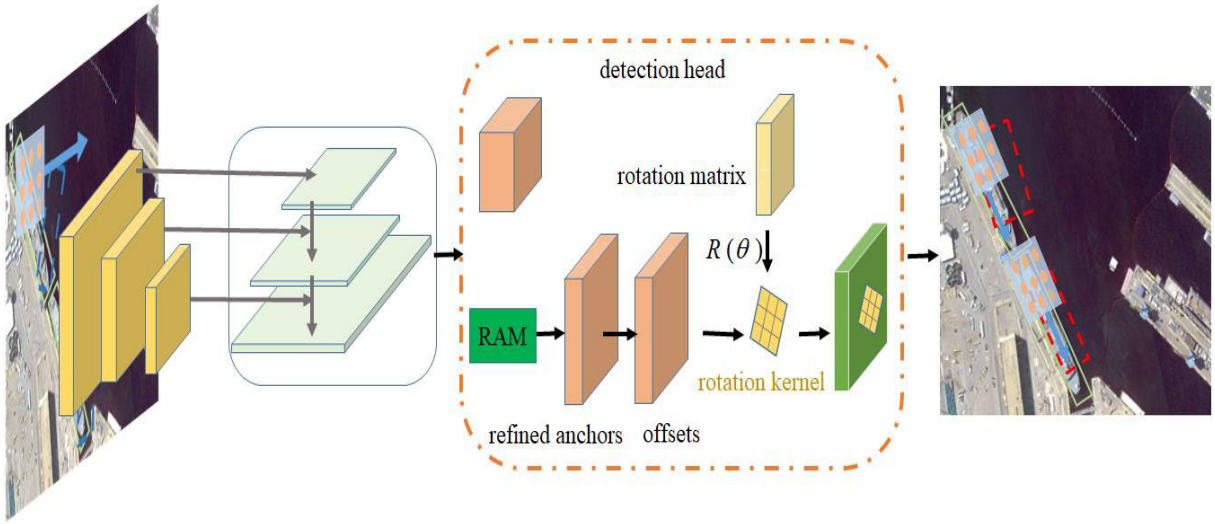


Fig. 7. Architecture of FAM. RRM makes up the detection head and refined anchors and the feature maps are fed into the FAM to extract aligned features.

TABLE III
EFFECTS OF EACH COMPONENT OF HAA-NET ON UCAS-AOD

with RRM ?	with FAM ?	with PLAM ?	mAP
✗	✗	✗	79.8%
✓	✓	✗	87.1%
✗	✓	✓	88.2%
✗	✗	✓	82.9%
✓	✓	✓	**91.5%**

where σ means sigmoid function; I_{in} is equal to input IoU; I_{out} represents the feature alignment capability, its value is the IoU between the prediction result of a box and the corresponding GT, which is the output IoU. ε, μ are two superparameters. $\sigma(I_{\text{out}} - I_{\text{in}})$ is used to represent the regression uncertainty, and it is obtained through the change between the output IoU and the input IoU. High-quality anchor sampling and stable training are very important to suppress interference in the regression process. When the difference between I_{out} and I_{in} is too large, the regression process produces a large gradient, which indicates that the anchor of the training process is quite unstable. The change of IoU represents the possibility of incorrectly sampled anchors.

Depending on the defined alignment degree, we can find negative samples with potential regression ability. In the training phase, we first calculate the alignment degree of GT and anchor and then select the anchor whose alignment degree is higher than the fixed threshold (set to 0.4 in our experiment) as the positive sample. Then, for the remaining GT that does not match the anchor, we choose the anchor with the largest alignment as the positive sample. Some of these anchors also have great regression ability after training, thus retaining the potentially high-quality negative samples.

D. Joint Optimization Loss Function

We jointly optimize all components to realize the hierarchical alignment of the network. The optimization of detectors is essentially a multitask learning process. The weight between tasks is adjusted by multiplying a fixed coefficient. Generally, the loss function in object detection is defined as follows:

$$L_{p_c,t_c,p_c^r,t_c^r} = L_{\text{cls}}(p_c, t_c) + \lambda[c \geq 1]L_{\text{reg}}\left(p_c^r, t_c^r\right) \tag{8}$$

where L_{cls} and L_{reg} are the loss functions of the classification task and regression task, respectively. And p_c and t_c represent the classification prediction value and category label of the

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network, respectively. In the regression loss function, p_c^r and t_c^r represent the regression result corresponding to the predicted classification result and the corresponding regression label. When $u = 0$ (negative sample), the regression loss is not calculated. Moreover, λ can be used as a parameter to balance classification and regression tasks.

To effectively establish the correlation of classification and regression tasks in object detection of arbitrary orientations, the joint optimization loss function based on the gradient equalization strategy is proposed to ensure the stable training of the model. For the classification branch, we design the matching degree loss function. By integrating the matching degree in PLAM, we can effectively screen the positive samples with regression potential. For the regression branch, we design the gradient equalization loss function, which makes the model pay more equal attention to all kinds of samples by suppressing the adverse impact of the large gradient generated by the target with any direction and large scale in the training process. Finally, we add the loss function of RRM to the joint optimization. In this way, the more stable training of the detector is guaranteed.

1) Alignment Loss: To effectively establish the correlation of classification and regression tasks in object detection of arbitrary orientations, the alignment degree is integrated into the training, and the alignment loss function is proposed

$$L_{\text{cls}} = \frac{1}{N} \sum w_i \cdot \text{FL}(p_c, t_c) \tag{9}$$

where w_i is a matching compensation factor for the candidate positive sample i. It is used to distinguish positive samples with different regression capabilities. FL is focal loss. This function can reduce the weight of easily classified samples so that the model can focus more on the samples that are difficult to classify during training. For the respective GT, its alignment degree with all anchors is first calculated and put into set M, and then the candidate positive samples are selected in accordance with a fixed threshold expressed as ad_{pos}, where $\text{ad}_{\text{pos}} \subseteq M$. Assumed maximum matching degree of M is ad_{max}, and w_i can be defined as

$$w_i = \text{ad}_{\text{pos}} + 1 - \text{ad}_{\text{max}}. \tag{10}$$

In the detector, the weak correlation between classification and regression ability would lead to the selection of prediction results according to the classification confidence. Using the designed matching compensation factor, the network can distinguish positive samples with different regression capabilities. Thus, the network can make high-quality predictions with scores of the classification task, which can alleviate the inconsistency. After training with an alignment loss, a higher classification score accurately reflects the higher localization performance represented. And it can be further used to promote the regression of the model in network training.

2) Gradient Equalization Loss: In general, the balance training of multitasking can be realized by appropriately increasing the coefficient of the loss function in the regression task. However, in RS images, due to targets with large scale and arbitrary orientation, the regression task would produce a large number of points which would bring great loss and

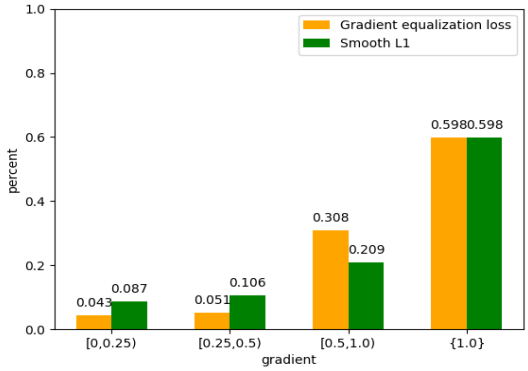


Fig. 8. Histogram of the statistical distribution of deviation point and near point based on HRSC2016. For the regression task of aerial targets, at least 60% of the samples contribute to the large gradient, while the samples that can produce a small gradient are ignored in the training.

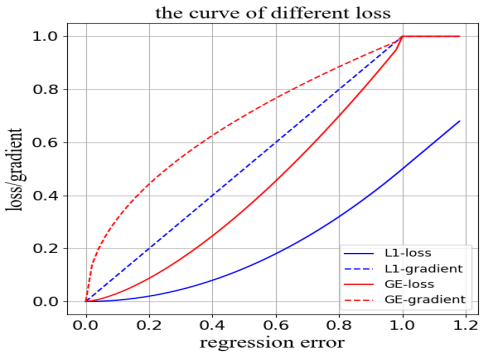


Fig. 9. Curves of different loss functions and corresponding gradients. The blue curve represents the L1 norm and the red curve represents our proposed gradient equalization loss function.

gradient. We call the deviation point when the regression loss is greater than or equal to 1.0 and the near point when the regression loss is less than 1.0. The distribution of deviation and the near point is shown in Fig. 8. Increasing the coefficient of the loss function in the regression task makes the model more inclined to the deviation point and ignores the near point. Therefore, in order to realize more stable training, we propose a gradient suppression function. L_{reg} in object detection is usually defined as

$$L_{\text{reg}}(x) = \text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{11}$$

The partial derivative of the above-mentioned formula can be obtained

$$\frac{\partial \text{smooth}_{L_1}}{\partial x} = \begin{cases} x, & \text{if } |x| < 1 \\ \pm 1, & \text{otherwise.} \end{cases} \tag{12}$$

As shown from this formula, when $x < 1$, the gradient increases linearly. Based on the above-mentioned equation, we design an improved gradient formula, which is defined as

$$\frac{\partial L_{\text{ge_reg}}}{\partial x} = \begin{cases} \alpha\sqrt{\beta|x|}, & \text{if } |x| < 1 \\ \gamma, & \text{otherwise} \end{cases} \tag{13}$$

where $L_{\text{ge_reg}}$ represents the loss function defined by us, and α is the gradient equalization factor. The gradient of the near

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TABLE IV
ABLATION STUDY OF THE PROPOSED RRM ON HRSC2016

refinement stages	0	1	2	3
mAP	88.7	**90.4**	87.4	83.2

TABLE V
ABLATION STUDY OF THE PROPOSED LOSS FUNCTION ON HRSC2016 DATASET

Parameter settings	$\lambda_1 = 0.5$		
	$\lambda_2 = 0.5$	$\lambda_2 = 1.0$	$\lambda_2 = 1.5$
mAP	**90.4**	89.3	86.3

TABLE VI
ABLATION STUDY OF THE PROPOSED FAM ON HRSC2016 DATASET

method	Conv	DeformConv	GA-DeformConv	FAM(ours)
mAP	83.3	84.2	84.7	**85.3**

TABLE VII
ABLATION STUDY OF THE PROPOSED PLAM ON HRSC2016 DATASET

Parameter settings	$\mu = 4$			$\mu = 5$		
	$\varepsilon = 0$	$\varepsilon = 0.3$	$\varepsilon = 0.5$	$\varepsilon = 0$	$\varepsilon = 0.5$	$\varepsilon = 0.7$
mAP	89.8	90.0	89.9	87.5	**90.4**	86.3

TABLE VIII
ABLATION STUDY OF THE PROPOSED GRADIENT EQUALIZATION LOSS VALUATION ON HRSC2016 DATASET

Parameter settings	$\alpha = 0.3$ $\gamma = 1.0$	$\alpha = 0.5$ $\gamma = 1.0$	$\alpha = 0.3$ $\gamma = 1.5$	$\alpha = 1.0$ $\gamma = 1.5$
mAP	83.8	89.3	88.5	**90.4**

TABLE IX
DETECTION RESULTS ON UCAS-AOD DATASET

Methods	Car	Airplane	mAP
R-RetinaNet [26]	84.65	85.46	78.19
R^2PN [29]	76.74	88.66	78.63
YOLOV3	74.63	89.52	82.08
RoI-Trans [31]	88.02	90.02	89.02
S2ANet [60]	89.56	90.42	89.99
RIDet-O [61]	88.88	90.35	89.62
HAA-Net	87.67	96.16	**91.52**

point can increase more with a small increase of α. Meanwhile, it does not produce an effect on the deviation point. β is used to control the slope of the gradient. γ is used to ensure the continuity of the loss function. When increasing in a small range, our curve enables the network to obtain more gradients. By integrating the expression, the final expression of the gradient suppression loss function can be obtained

$$L_{\text{ge_reg}}(x) = \int \frac{\partial L_{\text{ln_reg}}}{\partial x} dx = \begin{cases} \frac{2\alpha}{3\beta}(\beta|x|)^{\frac{3}{2}}, & \text{if } |x| < 1 \\ \gamma|x| + C, & \text{otherwise} \end{cases} \quad (14)$$

where β can be calculated by the critical condition. Substitute the critical value $x = 1$ to obtain $2\alpha(\beta|)^{(3/2)}/(3\beta) = \gamma$, we can get $\beta = 9(\gamma/\alpha)^2/4$. The comparison between our loss function and smooth L1 loss is shown in Fig. 9. Overall, the multitask loss function for HAA-Net is a weighted sum of three losses

$$L = L_{\text{cls}}(p_c, t_c) + \lambda_1 L_{\text{ref}}\left(p_c{}^{\text{ref}_r}, t_c^r\right) + \lambda_2 L_{\text{ge_reg}}\left(p_c^r, t_c^r\right) \quad (15)$$

where L_{cls}, L_{ref}, and $L_{\text{ge_reg}}$ are the classification loss, anchor refinement loss, and regression loss, respectively. p_c and t_c represent the classification prediction and category label of the network respectively. $p_c{}^{\text{ref}_r}$ and p_c^r represent the refined anchor and the predicted box corresponding to the predicted classification result, respectively. And t_c^r is the corresponding real-bounding box.

IV. EXPERIMENTS

In this section, we conduct extensive experiments on three aerial datasets, Dataset of Object Detection in Aerial Images (DOTA), the University of Chinese Academy of Sciences-Aerial Object Detection (UCAS-AOD), and High-Resolution Ship Collection 2016 (HRSC2016), and compare the proposed method with other relevant existing advanced detectors to prove the effectiveness of the hierarchical adaptive alignment network (HAA-Net).

A. Datasets

DOTA [46] is the largest public dataset for target detection based on RS images. DOTA includes 2806 aerial images and 188 282 annotation examples. There are a total of 15 categories, and the information for each category is shown in Table I. The visual detection results for each category are shown in Fig. 10. It should be noted that the size of the image in DOTA is too large. We cut the original image into 800 × 800 pieces. Random horizontal flips are used in the training to avoid overmatching.

HRSC2016 [14] is a complex scene dataset for ship detection released in 2016 with a total of 1061 high-resolution pieces, including ocean g ships and port ships ranging in size from 300 × 300 to 1500 × 900. The whole dataset is collected from famous ports, covering the training set (436), validation set (181), and test set (444). For our all experiments, the image sizes are resized to 800 × 800.

UCAS-AOD [47] refers to a remote-sensing aircraft and vehicle dataset. There are 1510 images approximately from Google Earth. It contains two categories: 510 car images and 1000 aircraft images. Due to having no official artificial division of images, we divide all images into test sets, verification sets, and training sets, with a ratio of 5:2:3. In the experiment here, all image scales are resized to 800 × 800.

B. Implementation Details

1) Evaluation Metric: In target detection, targets with different positions and categories may appear in each picture, and the classification and location performance of the model needs to be evaluated. The standard evaluation metric in image classification cannot be directly used in the target detection task. Therefore, the measurement standard mean average precision (mAP) is proposed for the target detection

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Fig. 10. Visual detection results on DOTA. Boxes with different colors represent different categories.

task. Specifically, we first obtain the precision and recall of the target

$$\text{Precision} = \frac{\text{TP}}{\text{DTs}}$$
$$\text{Recall} = \frac{\text{TP}}{\text{GTs}} \tag{16}$$

where TP represents the real sample. According to the Pascal VOC measurement standard widely used in the object detection field, if the IoU is greater than 0.5, the predicted box is regarded as TP. DTs represent all positive samples predicted, and GTs represents all GT.

For an excellent detector, we hope that the precision and recall of the model are very high, so we need to comprehensively consider these two factors. Therefore, we can get the AP value by constructing the P–R curve and calculating the area under the curve

$$\text{AP} = \int_0^1 p_{\text{smooth}}(r)dr. \tag{17}$$

The area under the curve is understood as the average value of all accuracy under different recall values. The closer the area is to 1, the better the performance. An mAP is used to calculate the AP of all categories and the average value. Its definition is as follows:

$$\text{mAP} = \frac{\sum_{i=1}^{K} \text{AP}_i}{K} \tag{18}$$

where K represents the number of categories. Generally, the mAP is usually expressed as a percentage, and we omit the percentage symbol in the experimental results of this article.

Moreover, we use the new definition of precision and recall weighted by IoU to measure the alignment degree of the detector. IoU is usually used to indicate the degree of overlap between the prediction box and the GT. Therefore, we weigh the value of IoU into precision and recall. The new definition of precision and recall based on IoU is

$$\text{Precision}' = \frac{\Sigma(\text{tp}^*\text{IoU})}{\text{DTs}}, \quad \text{Recall}' = \frac{\Sigma(\text{tp}^*\text{IoU})}{\text{GTs}} \tag{19}$$

where tp represents the real sample, and it is equal to 1. DTs represents all positive samples predicted, and GTs represents all GT.

2) Parameter Setting: ResNet is adopted as the backbone of HAA-Net. For the original ResNet, the ImageNet is directly applied to pretrained models from PyTorch. The feature pyramid of $P_3 - P_7$ is used for multiscale detection. The threshold of the positive sample matching process is 0.4, while the confidence of the detection head is 0.6. In addition, when the momentum is 0.9, the proposed HAA-Net is optimized by the Adam optimizer. The initial learning rate is set to 0.001, and the attenuation rate is 0.1 after 30 000 iterations. We carried out all experiments on a server equipped with NVIDIA 1080ti based on the PyTorch framework. RetinaNet is a simple one-stage detector and is adopted as the baseline. Any module introduced can complicate the computation. Different settings and architectures are set on RetinaNet. According to Table II, RetinaNet has 83.3% mAP, which demonstrates that the baseline of this study is solid.

C. Ablation Studies

1) Evaluation on Different Components: We conduct relevant ablation experiments on HRSC2016 and UCAS-ADO to verify the performance of the proposed different modules. Table II lists the achieved results on HRSC2016. The baseline model achieves only 83.3% mAP since the manual anchor does not automatically align with the object at a different level. When using RRM, the performance of the detector is optimized by 4.1%, which shows that RRM can generate high-quality anchors to improve performance. In the case of adding FAM, it can adaptively align the spatial feature with arbitrary direction according to the corresponding anchor. Our FAM constructs strong feature representation ability, and the performance of our HAA-Net is improved by 5.4%. When PLAM is added, the model accuracy improves by 1.7%, which indicates that the correct assignment of labels eliminates the misalignment phenomenon and effectively guides

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TABLE X
PERFORMANCE EVALUATION ON HRSC2016 DATASET

Methods	Backbone	Size	NA	mAP
Two-stage:				
R^2CNN [25]	ResNet101	800×800	21	73.1
RRPN [48]	ResNet101	800×800	54	79.1
R^2PN [29]	VGG16	-	24	79.6
RoI-Trans [31]	ResNet101	512×800	5	86.2
Gliding Vertex [18]	ResNet101	512×800	5	88.2
One-stage:				
RRD [62]	VGG16	384×384	13	84.3
R^3Det [24]	ResNet101	800×800	21	89.3
OPLD [63]	ResNet101	800×800	9	88.4
BBAVectors [53]	ResNet101	800×800	9	89.7
R-RetinaNet [26]	ResNet101	800×800	121	89.2
AR2Det [64]	ResNet101	512×512	-	89.6
SDet [65]	ResNet101	800×800	27	89.2
HAA-Net	ResNet101	800×800	8	**90.4**

TABLE XI
DETECTION PERFORMANCE ON HRSC2016 FOR NEW EVALUATION METRICS

Methods	Backbone	Recall′	Precision′
R^3Det [19]	ResNet101	86.3	83.4
R-RetinaNet [26]	ResNet101	85.56	87.2
RSDet [54]	ResNet101	85.2	86.5
HAA-Net	ResNet101	88.3	**89.4**



Fig. 11. Comparison between the preset anchor and the refined anchor by RRM. The red boxes and green boxes indicate the preset anchor boxes and the refined anchors, respectively.

the stable training of the network. On UCAS-AOD, as shown in Table III, we can achieve similar experimental results.

Compared with only one module, the network with the superposition of different modules achieves better performance. The participation of FAM and PLAM brings extensive features and correct label guidance, thereby achieving better regression and classification results. Meanwhile, experiments also show that the proposed modules have no conflict. When all the proposed methods are employed, the model exhibits the optimal performance of 90.4% mAP.

2) Evaluation on RRM: Based on FAM and PLAM, experiments in Table IV are performed to test the effect of the number of RRM. We can achieve 88.7% mAP with the combination of FAM and PLAM. When one module is adopted, the performance is optimized by 1.7% mAP. However, with two RRMs, the network reduces performance by 3.0%. This is because RRM can provide a high-quality anchor when the number is 1. When the number is equal to 2, with the increase of the regression stage threshold, the positive sample which exceeds the matching threshold would decrease significantly, resulting in a serious imbalance during training. This imbalance is particularly evident when the refinement stage is equal to 3. Therefore, in our HAA-Net, one RRM is the best choice to achieve great performance. In addition, we conduct relevant experiments on the coefficients of the loss function. As shown in Table V, when $\lambda_1 = 0.5$, $\lambda_2 = 0.5$, the detector has the best performance. We think that this is because the regressor and RRM have the same importance for the detector.

Moreover, we visually compare the refined anchor with the preset rotated anchor. As shown in Fig. 11, the red box represents the preset anchor, and the green one represents the anchor refined by RRM. We can see that the refined anchor fits more closely with the target, which realizes the effective alignment between the anchor and the target.

3) Evaluation on FAM: This study compares FAM with other convolutions to verify its effectiveness. This study only adopts other convolution methods to replace FAM and keeps other settings unchanged. FAM reconstructs the feature map and increases the overall performance by nearly 2% compared with the standard convolution. According to Table VI, FAM is significantly better than other methods. At the same time, the mAP of DeformConv and GA-DeformConv only reach 84.2% and 84.7%. As revealed by the experiment results, FAM reduces the number of targets that are underfitting due to the misalignment features and achieves accurate regression results when detecting dense and arbitrary targets.

4) Evaluation on PLAM: To find superparameters suitable for PLAM and explore the correlation between parameters, a parameter sensitivity experiment is conducted, and the results are analyzed in Table VII. With the appropriate reduction of ε, the influence of feature alignment increases, which shows that the output IoU helps select anchors with high positioning accuracy.

However, when ε is significantly large ($\varepsilon = 0.7$), the network performance declines sharply. This is because most of the potential high-quality samples are effectively suppressed when the output IoU is difficult to present feedback information. Moreover, as shown in Table VIII, in our experiments, we can achieve the best result when setting $\alpha = 1.0$, $\gamma = 1.5$.

D. Comparisons With the State-of-the-Art

1) Results on DOTA: On the DOTA dataset, we compare our HAA-Net with other latest methods. As shown in Table I, for plane (PL), bridge (BR), ground track field (GTF), tennis court (TC), basketball court (BC), soccer-ball field (SBF), roundabout (RA), our AP obtained the best results: 90.43%, 62.23%, 84.71%, 90.90%, 90.50%, 69.77%, and 84.54%, respectively. In addition, we still obtain the best average results for all categories with 79.40% mAP. Some visual detection results of aerial targets on DOTA are shown in Fig. 10. Our HAA-Net can accurately detect densely arranged small objects

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Fig. 12. Visual detection results on UCAS-AOD.

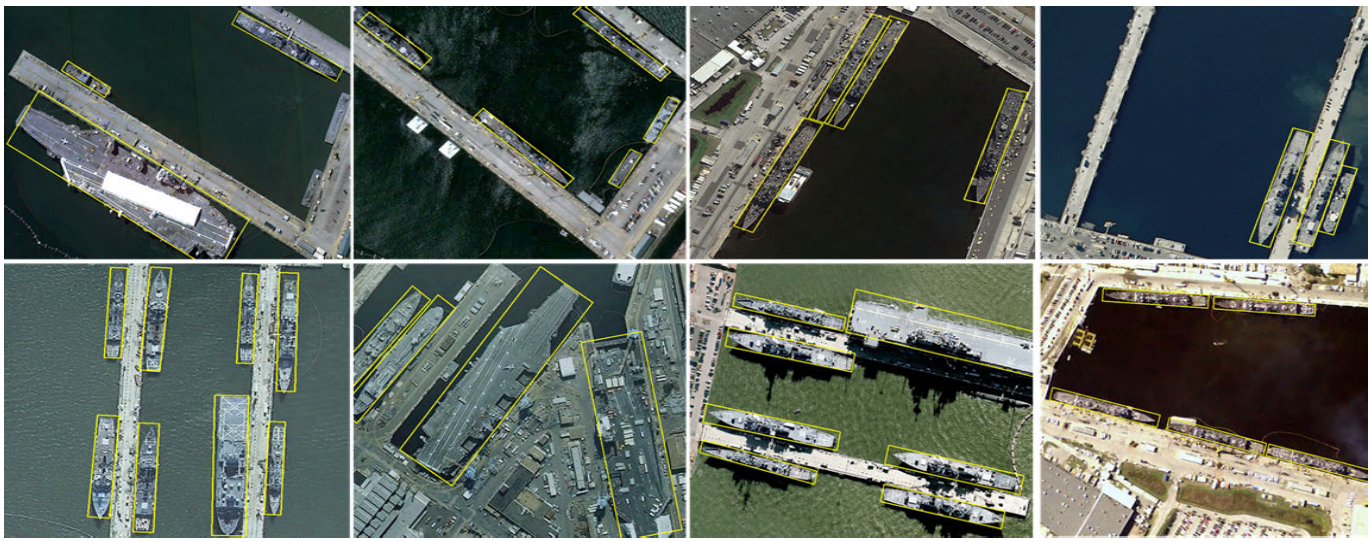


Fig. 13. Visual detection results on HRSC2016 dataset.

(such as small vehicles). For targets with arbitrary orientation (such as planes and ships), HAA-Net can accurately align the spatial direction of these targets to realize the adaptation to rotation. In addition, the detection network can well adapt to scale changes and accurately locate targets with different scales. And our HAA-Net realizes the accurate detection of large-scale roundabouts and small vehicles at different scales. In addition, our method can use some square anchors to detect objects with a very large aspect ratio (such as bridges and ports here).

These results show that hierarchical adaptive alignment for the aerial object is necessary. The key to achieving accurate detection is the effective alignment of region level, feature level, and label level. RRM corrects the preset box to ensure a high-quality sample, FAM fully extracts the target spatial features to ensure the quality of the features, and PLAM guarantees the accurate assignment of labels to facilitate the stable training of the network. The three modules work together to ensure network performance.

2) Results on UCAS-AOD: According to the obtained detection results in Table IX, our HAA-Net shows the best performance, reaching 91.52% of mAP among the existing methods, including two-stage and one-stage detectors. In addition, the objects in UCAS-AOD have large-scale changes, arbitrary directions, and dense distribution. The excellent performance shows the robustness of dense target detection, which proves that the proposed method can align the anchor with target features and effectively distinguish dense targets to achieve high-quality detection. And our method effectively eliminates the error of a heuristic-defined anchor. Qualitative results of UCAS-AOD are shown in Fig. 12.

3) Results on HRSC2016: This dataset contains many types of ships moored in ports and oceans. We compare HAA-Net with other existing detectors in Table X. "NA" indicates the number of anchors. By using the proposed modules, our HAA-Net achieves excellent performance with 90.4% mAP. Compared with the specific ship detectors, our HAA Net is 0.8% and 1.2% mAP higher than AR2Det and SDet, respec-

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Fig. 14. Comparison of alignment of different detectors. The red box and green box indicate the prediction result of HAA-Net and SCRDet++, respectively, and the blue box is GT.

tively. The performance of HAA-Net with fewer anchors is better than other advanced one-stage and two-stage detectors. Qualitative results of HAA-Net are visualized in Fig. 13. Moreover, the feature level used in the feature pyramid is $P_3 - P_7$. Compared with two-stage detectors with $P_2 - P_6$, the total amount of positions required to preset the anchor is less.

We compare several advanced detectors with our method on HRSC2016 and calculate IoU between the prediction box and the GT. In Fig. 14, the yellow box is predicted by HAA-Net, the red box is predicted by SCRDet++, and the green box is GT. It can be seen from Fig. 14 that although our HAA-Net and SCRDet++ have detected the target, our prediction box has a higher IoU. This can prove that our detection box fits the target more tightly.

4) Results on New Evaluation Metrics: We compare HAA-Net with other methods on HRSC2016 by using new evaluation metrics. As can be seen from Table XI, compared with R^3Det, R-RetinaNet, and RSDet by using the new definition, we still achieve the best result in terms of Recall′ and Precision′, which is (2.0%, 2.8%, and 3.1%) and (6.0%, 2.2%, and 2.9%) higher than these detectors, respectively. The result further indicates the effectiveness of our method.

V. Conclusion

In this article, we systematically analyze the misalignment issues of model architectures for aerial object detection. To this end, we have proposed a novel HAA-Net which consists of three modules: RRM, FAM, and PLAM. On that basis, by using the gradient equalization strategy designed for different levels, our HAA-Net can complete multilevel alignment to improve detector performance. Extensive experiments on three public aerial benchmarks show the effectiveness of our method.

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